

回归分析 第一次上机作业 2112054 王赫明

***以下分析均以“Employed”作为响应变量。**

1.检查复共线性;

1.1 相关性分析

```
In [1]: library(corrplot)

#定义回归自变量，因变量
longley.x <- longley[1:6]
longley.y <- longley$Employed

#求相关系数阵（标准归一化）
cor_x <- cor(longley.x)
cor_x
corrplot(cor_x)
```

Warning message:
"package 'corrplot' was built under R version 4.2.3"
corrplot 0.92 loaded

A matrix: 6 × 6 of type dbl

	GNP.deflator	GNP	Unemployed	Armed.Forces	Population	Year
GNP.deflator	1.0000000	0.9915892	0.6206334	0.4647442	0.9791634	0.9911492
GNP	0.9915892	1.0000000	0.6042609	0.4464368	0.9910901	0.9952735
Unemployed	0.6206334	0.6042609	1.0000000	-0.1774206	0.6865515	0.6682566
Armed.Forces	0.4647442	0.4464368	-0.1774206	1.0000000	0.3644163	0.4172451
Population	0.9791634	0.9910901	0.6865515	0.3644163	1.0000000	0.9939528
Year	0.9911492	0.9952735	0.6682566	0.4172451	0.9939528	1.0000000



```
In [2]: #计算k值
kappa(cor_x, exact=TRUE)
```

12220.0098602718

计算得到的 $k=12220 \gg 1000$ 。以Employed为回归因变量时，具有严重的复共线性。

1.2 VIF (方差膨胀因子) 分析

```
In [3]: library(car)
vif(lm(Employed~., data=longley))
```

Warning message:
"package 'car' was built under R version 4.2.3"
Loading required package: carData

Warning message:
"package 'carData' was built under R version 4.2.3"

GNP.deflator: 135.532438280064 **GNP:** 1788.51348271901 **Unemployed:**
33.6188905960563 **Armed.Forces:** 3.58893019344596 **Population:** 399.151022312826
Year: 758.980597407239

除Unemployed和Armed.Forces的VIF值较小外，其余均 >100 ，可认为复共线性主要来源于这四个自变量。

2.使用主成分回归解决复共线性，选择适当个数的主成分；

```
In [4]: #进行主成分分析，观察各主成分贡献率，确定保留的主成分个数
pca<-princomp(longley.x, cor=TRUE)
summary(pca, loadings=TRUE)
```

Importance of components:

	Comp. 1	Comp. 2	Comp. 3	Comp. 4	Comp. 5
Standard deviation	2.1455482	1.0841312	0.45102702	0.122181253	0.0505179747
Proportion of Variance	0.7672295	0.1958901	0.03390423	0.002488043	0.0004253443
Cumulative Proportion	0.7672295	0.9631196	0.99702383	0.999511871	0.9999372153

	Comp. 6
Standard deviation	1.940897e-02
Proportion of Variance	6.278469e-05
Cumulative Proportion	1.000000e+00

Loadings:

	Comp. 1	Comp. 2	Comp. 3	Comp. 4	Comp. 5	Comp. 6
GNP.deflator	0.462		0.149	0.793	0.338	0.135
GNP	0.462		0.278	-0.122	-0.150	-0.818
Unemployed	0.321	-0.596	-0.728			-0.107
Armed.Forces	0.202	0.798	-0.562			
Population	0.462		0.196	-0.590	0.549	0.312
Year	0.465		0.128		-0.750	0.450

观察累积贡献率，在comp3已经到达99%。因此保留前三个主成分！

$F_1 = 0.462 \text{GNP.deflator} + 0.462 \text{GNP} + 0.321 \text{Unemployed} + 0.202 \text{Armed.Forces} + 0.462 \text{Population} + 0.465 \text{Year}$

$F_2 = -0.596 \text{Unemployed} + 0.798 \text{Armed.Forces}$

$F_3 = 0.149 \text{GNP.deflator} + 0.278 \text{GNP} + 0.728 \text{Unemployed} + -0.562 \text{Armed.Forces} + 0.196 \text{Population} + 0.128 \text{Year}$

```
In [5]: #定义每个主成分中原变量权重
weight_1 <- c(0.462, 0.462, 0.321, 0.202, 0.462, 0.465)
weight_2 <- c(0, 0, -0.596, 0.798, 0, 0)
weight_3 <- c(0.149, 0.278, -0.728, -0.562, 0.196, 0.128)

#计算主成分值，填入longley中作为新的回归变量
F_1<-c(0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)
F_2<-c(0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)
F_3<-c(0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)
for (i in 1:16) {
  F_1[i]<-rowSums(weight_1*longley.x[i,])
  F_2[i]<-rowSums(weight_2*longley.x[i,])
  F_3[i]<-rowSums(weight_3*longley.x[i,])
}
longley<-cbind(longley, F_1, F_2, F_3)
head(longley)
```

A data.frame: 6 × 10

	GNP.deflator	GNP	Unemployed	Armed.Forces	Population	Year	Employed	
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<int>	<dbl>	<
1947	83.0	234.289	235.6	159.0	107.608	1947	60.323	1209
1948	88.5	259.426	232.5	145.6	108.632	1948	61.122	1220
1949	88.2	258.054	368.2	161.6	109.773	1949	60.171	1267
1950	89.5	284.599	335.1	165.0	110.929	1950	61.187	1271
1951	96.2	328.975	209.9	309.9	112.075	1951	63.221	1285
1952	98.1	346.999	193.2	359.4	113.270	1952	63.639	1300

```
In [6]: #选择三个主成分作为回归自变量，进行拟合
lm.pca<-lm(Employed~F_1+F_2+F_3,data=longley)
summary(lm.pca)
```

Call:

```
lm(formula = Employed ~ F_1 + F_2 + F_3, data = longley)
```

Residuals:

```
      Min       1Q   Median       3Q      Max
-0.78807 -0.23269 -0.00027  0.12089  1.08484
```

Coefficients:

```
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -8.620557    2.940117  -2.932   0.0126 *
F_1           0.054530    0.002153  25.332 8.70e-12 ***
F_2           0.003248    0.001556   2.088   0.0588 .
F_3           0.033406    0.003747   8.915 1.22e-06 ***
---

```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Residual standard error: 0.4751 on 12 degrees of freedom

Multiple R-squared: 0.9854, Adjusted R-squared: 0.9817

F-statistic: 269.2 on 3 and 12 DF, p-value: 2.869e-11

确定回归方程为： $\text{Employed} = -8.620557 + 0.054530F_1 + 0.003248F_2 + 0.033406F_3$

```
In [7]: #验证PCA下新模型的复共线性  
longley_pca<-longley[8:10]  
kappa(cor(longley_pca))
```

7.15527587041799

k值降到了10以下，说明主成分回归极有效地解决了原自变量的复共线性！

3 岭回归

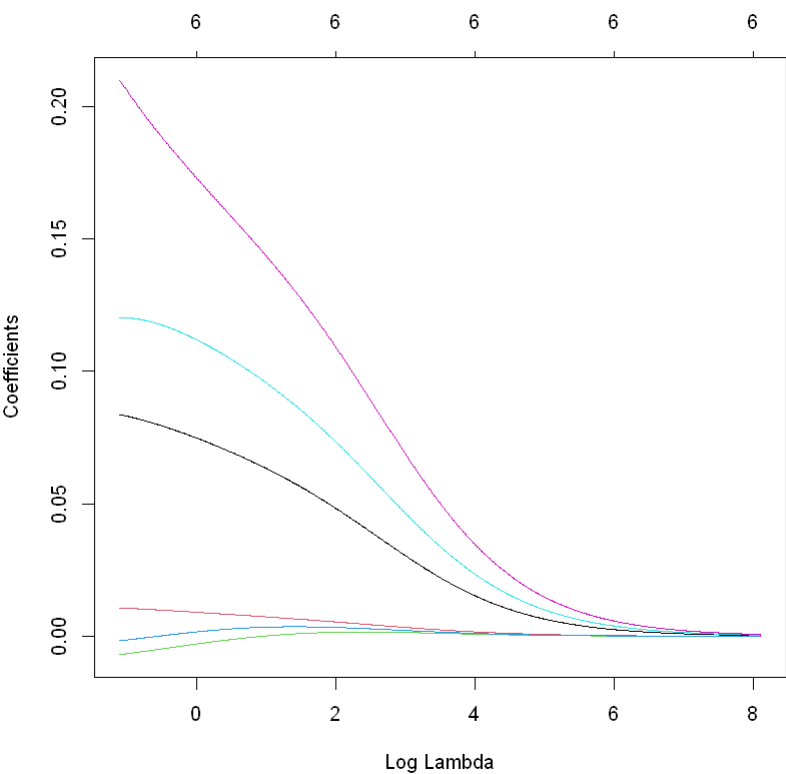
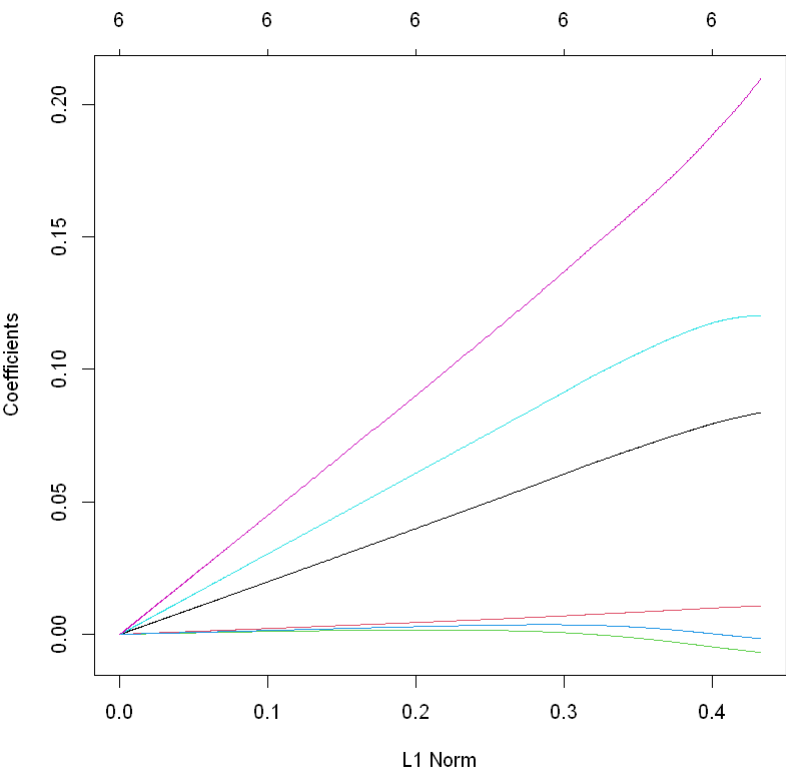
3.1 使用弹性网络方法 (glmnet包)

```
In [16]: #调取所需包
rm(list=ls())
library(glmnet)
library(readxl)
library(plyr)
library(corrplot)
library(ggplot2)

x <- as.matrix(longley[1:6])
y <- as.double(longley$Employed)
fit <- glmnet(x, y, family="gaussian", nlambda=1000, alpha=0)

plot(fit, lable=TRUE)
plot(fit, xvar="lambda", lable=TRUE)
```

```
Warning message in plot.window(...):
"lable" is not a graphical parameter"
Warning message in plot.xy(xy, type, ...):
"lable" is not a graphical parameter"
Warning message in axis(side = side, at = at, labels = labels, ...):
"lable" is not a graphical parameter"
Warning message in axis(side = side, at = at, labels = labels, ...):
"lable" is not a graphical parameter"
Warning message in box(...):
"lable" is not a graphical parameter"
Warning message in title(...):
"lable" is not a graphical parameter"
Warning message in plot.window(...):
"lable" is not a graphical parameter"
Warning message in plot.xy(xy, type, ...):
"lable" is not a graphical parameter"
Warning message in axis(side = side, at = at, labels = labels, ...):
"lable" is not a graphical parameter"
Warning message in axis(side = side, at = at, labels = labels, ...):
"lable" is not a graphical parameter"
Warning message in box(...):
"lable" is not a graphical parameter"
Warning message in title(...):
"lable" is not a graphical parameter"
```




```
In [17]: ridge_fit <- cv.glmnet(x, y, family = "gaussian", type.measure = "auc", alpha = 0, nlambda = 100)
plot(ridge_fit)
ridge_fit$lambda.min
```

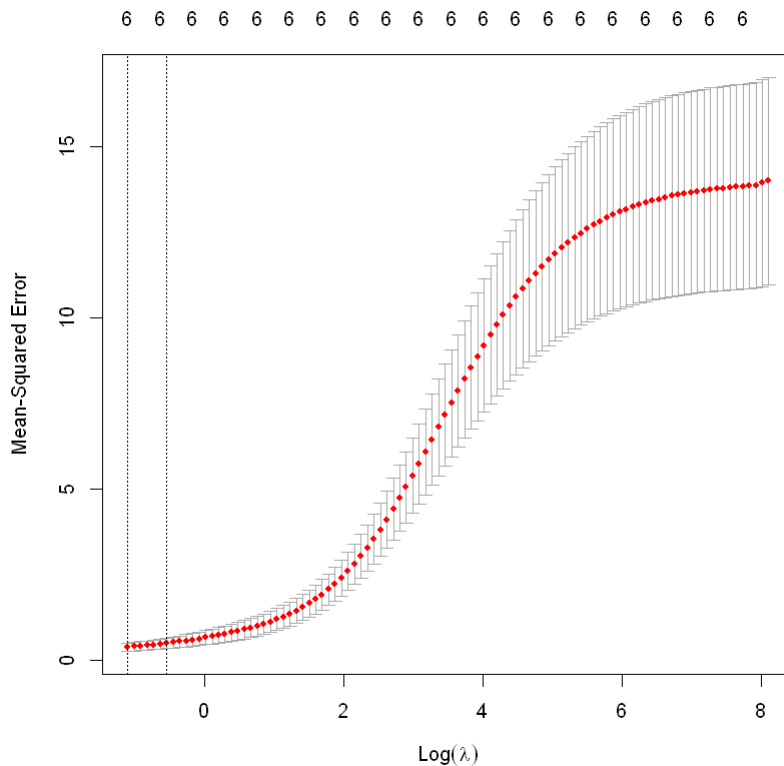
Warning message:

"Only mse, deviance, mae available as type.measure for Gaussian models; mse used instead"

Warning message:

"Option grouped=FALSE enforced in cv.glmnet, since < 3 observations per fold"

0.334451683575075



运用该方法得到的岭参数 k 为0.334451683575075。（代码参考了网上的一些范例，有的地方没有完全理解，所以基本没有作注释）

3.2 使用岭迹图

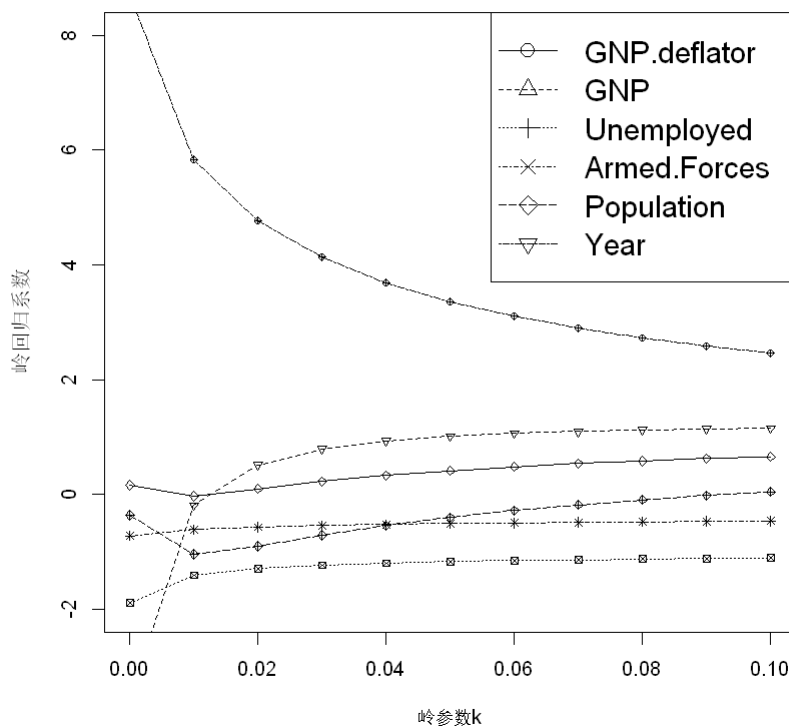
```
In [55]: datas<-data.frame(scale(longley[,1:6]))
library(MASS)
ridge<-lm.ridge(y~.-1,datas,lambda=seq(0,0.1,0.01))

# 做岭回归，对于标准化后的数据模型不包含截距项，其中lambda为岭参数k的所有取值
beta<-coef(ridge) # 将所有不同岭参数所对应的回归系数的结果赋给beta

# 绘制岭迹图
k<-ridge$lambda #将所有岭参数赋给k
plot(k,k,type="n",xlab="岭参数k",ylab="岭回归系数",ylim=c(-2,8))

# 创建没有任何点和线的图形区域
linetype<-c(1:6)
char<-c(5:10)
for(i in 1:6)
  lines(k,beta[,i],type="o",lty=linetype[i],pch=char[i],cex=0.75)

legend("topright", inset=0, legend=c("GNP.deflator","GNP","Unemployed","Armed.Forces",
"Population","Year"))
```



可以看到，在 $k=0.07$ 时，岭迹图已趋于平缓，故可将 k 的估计值定为0.07。

使用R中自带的select来估计岭参数 k

```
In [56]: ridge_select=lm.ridge(Employed~., longley, lambda=seq(0, 0.1, 0.0001))  
select(ridge_select)
```

```
modified HKB estimator is 0.004275357  
modified L-W estimator is 0.03229531  
smallest value of GCV at 0.0028
```